Polynomial Filter Diagonalization and Quantum Many-Body Scars in the PXP Model

Ahmet Berk Kavruk
University of Bonn
(Dated: June 19, 2026)

There will be some abstract here...

Lecture: physics767: Computational Methods in Condensed Matter Theory, Prof. Dr. David J. Luitz
Tutors: Kriti Baweja and Dr. Matías Gonzalez

I. INTRODUCTION

Why the topic is relevant and important, current research status etc.

II. THE PXP HAMILTONIAN

A. The Action of the PXP Hamiltonian

To study the Rydberg blockade mechanism, we introduce the PXP Hamiltonian

$$H = \sum_{i=0}^{L-1} P_i X_{(i+1)\%L} P_{(i+2)\%L}, \tag{1}$$

where $X_i = \mathbb{1} \otimes \sigma_x \otimes \mathbb{1}$ is the Pauli matrix $X = \sigma_x$, acting on site i, and the projector $P_i = \frac{1}{2}(\mathbb{1} - Z_i)$ is described in terms of the identity $\mathbb{1}$ and the Pauli matrix $Z = \sigma_z$, which acts on site i. The Hilbert space $\mathcal{H}$ is defined such that $Z|0\rangle = -|0\rangle$ and $Z|1\rangle = |1\rangle$, such that $P|1\rangle = 0$ and $P|0\rangle = |0\rangle$. For a computational basis state

$$|n_0 n_1 \cdots n_{L-1}\rangle, \qquad n_i \in \{0, 1\},$$

the Hamiltonian acts by checking whether the two neighboring sites i and $i+2$ are both in the state $|0\rangle$. If they are, the operator flips the middle spin at site $i+1$, if they are not, the term gives zero. A term in the Hamiltonian can be seen explicitly as,

$$P_i X_{(i+1)\%L} P_{(i+2)\%L} |\cdots n_i n_{i+1} n_{i+2} \cdots\rangle = \begin{cases} |\cdots 0(n_{i+1} \oplus 1)0 \cdots\rangle, & n_i = 0,\ n_{i+2} = 0, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Here, the $\oplus$ in Equation (2) is used to emphasize that the summation is in modulo 2.

To see the action explicitly, let us consider the local three-site basis

$$|n_0 n_1 n_2\rangle, \qquad n_i \in \{0, 1\}.$$

If we take one of the local Hamiltonian terms $h_0 = P_0 X_1 P_2$, we can see that h_0 acts only when both neighboring sites of n_1 are in the state $|0\rangle$. Therefore, it maps two of the basis states as

$$h_0 |000\rangle = |010\rangle,$$
$$h_0 |010\rangle = |000\rangle,$$

while the rest of the $h_0 |n_0 n_1 n_2\rangle = 0$. Thus, a spin can only flip if both of its neighboring sites are in the state $|0\rangle$. This effectively models the blockade constraint.

B. Blockade Constrained Hilbert Space

The Rydberg blockade constraint forbids both neighboring atoms to be in the excited state at the same time. This means the states that contain adjacent $|\cdots 11 \cdots\rangle$ configurations are not physically relevant. If we look at the adjacency graph of the Hamiltonian, which is constructed by connecting two basis states whenever the corresponding Hamiltonian matrix element $\langle n_0 n_1 \cdots | H | n_0' n_1' \cdots\rangle$ is nonzero, we see that the largest connected component is formed by the basis states that satisfy the Rydberg blockade constraint. Therefore, we restrict ourselves to the biggest block of the Hamiltonian for our calculations.

Another curious property of the biggest block is its dimension. Assume we have a chain of atoms of size L, and we have open boundary conditions, then we have the

Hilbert space of the biggest block with dimension

$$\dim \mathcal{H}_{\text{biggest}}(L) = \mathcal{D}(L), \tag{3}$$

and we add one more atom to our basis state. An atom has the states $|0\rangle$ and $|1\rangle$. The state $|0\rangle$ of the new atom can be freely attached to any biggest block basis state in the L-atom chain, giving us $\mathcal{D}(L)$ number of valid configurations. When adding the state $|1\rangle$ of the new atom to the L-atom chain, the Lth atom must be in the state $|0\rangle$, and we can again attach this pair $|0\rangle_L |1\rangle_{L+1}$ freely to any biggest block basis state in the remaining $(L-1)$-atom chain, giving us $\mathcal{D}(L-1)$ number of valid configurations. Therefore, the new $(L+1)$-atom chain has the Hilbert space dimension

$$\mathcal{D}(L+1) = \mathcal{D}(L) + \mathcal{D}(L-1), \tag{4}$$

which implies a Fibonacci sequence, with the initial conditions $\mathcal{D}(1) = 2$ and $\mathcal{D}(2) = 3$. This also implies that the dimension of the biggest block Hilbert space $\mathcal{D}(L)$ of the PXP Hamiltonian grows with the golden ratio

$$\phi^L = \left(\frac{1+\sqrt{5}}{2}\right)^L \approx (1.618)^L,$$

with growing system sizes, rather than 2^L.

For periodic boundary conditions, the blockade constraint also acts between the first and last atoms, which changes the counting from the open-chain case. This time, we can split the allowed configurations according to the value of the first site. If $|n_0\rangle = |0\rangle$, the remaining $L-1$ sites form an open chain with the blockade constraint, giving $\mathcal{D}_{\text{open}}(L-1)$ configurations. If $|n_0\rangle = |1\rangle$, then periodicity forces both neighboring sites to be zero, $|n_1\rangle = |n_{L-1}\rangle = |0\rangle$. The remaining free part can then be thought of as an $(L-3)$-atom chain with open boundary conditions, which has $\mathcal{D}_{\text{open}}(L-3)$ configurations. Thus we have

$$\mathcal{D}_{\text{periodic}}(L) = \mathcal{D}_{\text{open}}(L-1) + \mathcal{D}_{\text{open}}(L-3). \tag{5}$$

Using $\mathcal{D}_{\text{open}}(L) = F_{L+2}$, Equation (5) becomes

$$\mathcal{D}_{\text{periodic}}(L) = F_{L+1} + F_{L-1}, \tag{6}$$

which is the L-th Lucas number [1]. The constrained Hilbert space still grows asymptotically as ϕ^L, but with the periodic boundary condition included exactly.

C. Translation Invariance and Momentum Sectors

The PXP Hamiltonian has translation symmetry, since every site sees the same rule and the chain is periodic. We can use this to create a basis of symmetrized momentum states and block diagonalize the biggest block in momentum sectors with the moment k. The construction of the momentum basis in this section follows the lecture notes [2].

To use the translation invariance in our favor, we define the translation operator T as the operator which shifts every spin by one site,

$$T\,|n_0 n_1 \cdots n_{L-1}\rangle = |n_{L-1} n_0 \cdots n_{L-2}\rangle\,. \tag{7}$$

Since the Hamiltonian is a sum over identical local terms on every site, translation does not change the Hamiltonian. Therefore,

$$[H, T] = 0.$$

This means we can diagonalize H and T simultaneously, and the Hamiltonian can be decomposed into momentum sectors with the basis of translation invariant momentum states

$$|\psi_{k,\vec{n}}\rangle = \frac{1}{\mathcal{N}_{k,\vec{n}}} \sum_{r=1}^{L} e^{ikr} T^r\,|\text{rep}(\vec{n})\rangle\,. \tag{8}$$

Here $k = 2\pi n_k/L$ are the momentum sectors, $n_k \in \mathbb{Z}_L$, and $|\text{rep}(\vec{n})\rangle$ are representatives of different families (orbits) $\mathcal{F}(\vec{n})$, created by the translation operator T, defined as

$$|\text{rep}(\vec{n})\rangle = T^{x_{\vec{n}}}\,|\vec{n}\rangle\,. \tag{9}$$

Here, we use $|\vec{n}\rangle = |n_0 n_1 \cdots n_{L-1}\rangle$ as a shorthand notation. The representative $|\text{rep}(\vec{n})\rangle$ of a family is usually chosen as the index that has the lowest binary representation.

When we look at the normalization of Equation (8) for $f_{\vec{n}} = \#\mathcal{F}(\vec{n}) < L$, the normalization condition gives us a selection rule

$$f_{\vec{n}} n_k \pmod L = 0, \tag{10}$$

because the momentum projection is not always normalizable. If this rule is not satisfied, then the projected momentum state must be omitted from the basis. For normalizable states, we can write

$$|\psi_{k,\vec{n}}\rangle = \frac{1}{\sqrt{f_{\vec{n}}}} \sum_{r=1}^{f_{\vec{n}}} e^{ikr} T^r\,|\text{rep}(\vec{n})\rangle\,. \tag{11}$$

III. NUMERICAL METHODS

A. Sparse Hamiltonian and Adjacency Graph

A sparse matrix is a matrix where there are much more zero elements than nonzero elements. We can conveniently store the information of such matrices by only considering their nonzero elements. The matrix form of the PXP Hamiltonian is also sparse (refer to Equation (A1) for an example for $L = 4$), and we can find its elements by the algorithm,

1. Generate the computational basis states $\{|i\rangle\}$.
2. Initialize empty lists for the row indices, column indices, and matrix entries.
3. For each basis state $|i\rangle$ and each lattice site n, apply the local operator $P_{n-1}X_nP_{n+1}$.
4. If the action is allowed, identify the resulting state $|j\rangle$. We now have $\langle j|\,H\,|i\rangle = H_{ji}$.
5. Append j to the row list, i to the column list and a 1 to the data list.
6. Construct the sparse Hamiltonian by feeding the lists to `scipy.sparse.csr_matrix`.

After this, the adjacency graph can be conveniently drawn using `networkx.from_scipy_sparse_array`. The examples for full matrix and its adjacency graph can be seen in Equation (A1) and Figure 2 respectively.

B. Momentum Sector Basis Construction

C. Exact Diagonalization Benchmark

To be used as benchmark, we fully diagonalize the biggest block of the PXP Hamiltonian by using the dense matrix version of the Hamiltonian. Different blocks of the Hamiltonian can be accessed by using `networkx.connected_components` after forming the adjacency matrix. The biggest block of the Hamiltonian is then accessed by taking the list which has the longest length in a list of lists we get from `networkx.connected_components`. The exact diagonalization is then carried out by `scipy.linalg.eigh` after converting the sparse matrix to array. The result is then stored to be used as benchmark. The full dense spectrum for $L = 16$ case can be seen in Figure 1.

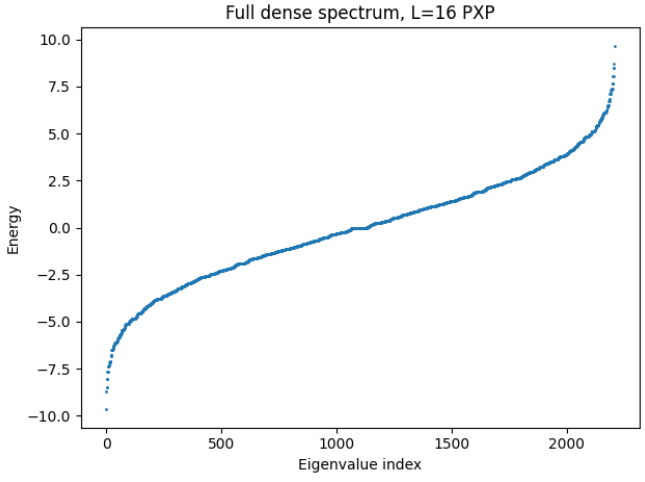


FIG. 1. Exact diagonalization for $L = 16$.

D. Entanglement Entropy via Reshaping

The scar states are identified by their unusually low entropy compared to usual thermal eigenstates at similar energies. Thus, comparing the entanglement entropy against the energy is a convenient way to identify the scar states inside the spectrum. The computation of the entanglement entropy follows the script by Ulrich Schollwöck [3].

To compute the entanglement entropy, the system is thought of as a bipartite system. For a chain of length L, we divide the system into two subsystems A and B, with lengths L_A and L_B, such that $L_A + L_B = L$. A general eigenstate can be written as

$$|\psi\rangle = \sum_{i_A,i_B} \psi_{i_Ai_B}\,|i_A\rangle \otimes |i_B\rangle\,, \tag{12}$$

where $|i_A\rangle$ and $|i_B\rangle$ are computational basis states of the two subsystems. The coefficients of the state are then reshaped into a matrix

$$M_{i_Ai_B} = \psi_{i_Ai_B}. \tag{13}$$

The singular values of this matrix are the Schmidt coefficients of the bipartition. Equivalently, if

$$M = USV^\dagger, \tag{14}$$

then the diagonal entries s_α of S give the Schmidt coefficients.

The reduced density matrix of subsystem A, defined by $\rho_A = \mathrm{Tr}_B(\rho)$, has eigenvalues

$$\lambda_\alpha = s_\alpha^2, \tag{15}$$

and the von Neumann entanglement entropy is

$$S_A = -\sum_\alpha \lambda_\alpha \log \lambda_\alpha = -\sum_\alpha s_\alpha^2 \log s_\alpha^2. \tag{16}$$

In the numerical implementation, this can be directly computed by reshaping the eigenvector into a $2^{L/2}\times 2^{L/2}$ matrix, performing a singular value decomposition, and summing over the squared singular values. However, because of the floating point arithmetic, Schmidt values can get absurdly low, causing `numpy` to interpret them as zero inside the logarithm. Therefore a tolerance of $s_\alpha > 10^{-15}$ should be taken.

Since the Hamiltonian is diagonalized in the blockade constrained Hilbert space, the eigenvectors first have to be embedded back into the full computational basis before reshaping to compute their entropies correctly.

E. Chebyshev Polynomial Filtering

The scar states live on a broad spectrum, thus if we want to find them we need to be specifically targeting

them by using a filtering method that amplifies the targeted energy eigenstate's eigenvalue and suppresses the rest. In theory, this can be done by using the Dirac delta function

$$\delta(E - H), \tag{17}$$

which would project onto eigenstates with energy E, so we would be able to target specific energy eigenstates directly. However, numerically this is not directly usable because the peak of Dirac delta function is infinitely sharp. Because of this, we replace the Dirac delta function with some approximation of it by expanding the delta function at the target value with some polynomial series, this time the Chebyshev polynomials.

Chebyshev polynomials are defined by

$$T_n(x) = \cos(n \arccos x), \tag{18}$$

with the inner product

$$\int_{-1}^{1} dx\, T_m(x) T_n(x) / \sqrt{1 - x^2} = \delta_{nm}, \tag{19}$$

and satisfy the recursion relation

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x). \tag{20}$$

Chebyshev polynomials are used because they are the optimal way to do polynomial filtering for Hermitian matrices, and their recurrence relation gives us a convenient way to calculate the current vector product with using the old vector products [2].

Appendix A: Dense Hamiltonian for $L = 4$

For illustration, the full Hamiltonian for $L = 4$ in the computational basis can be seen in Equation (A1).

$$H = \begin{pmatrix}
0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix} \tag{A1}$$

Appendix B: Adjacency Graph for $L = 4$

For illustration, the adjacency graph of thee Hamiltonian for $L = 4$ in the computational basis can be seen in Figure 2.

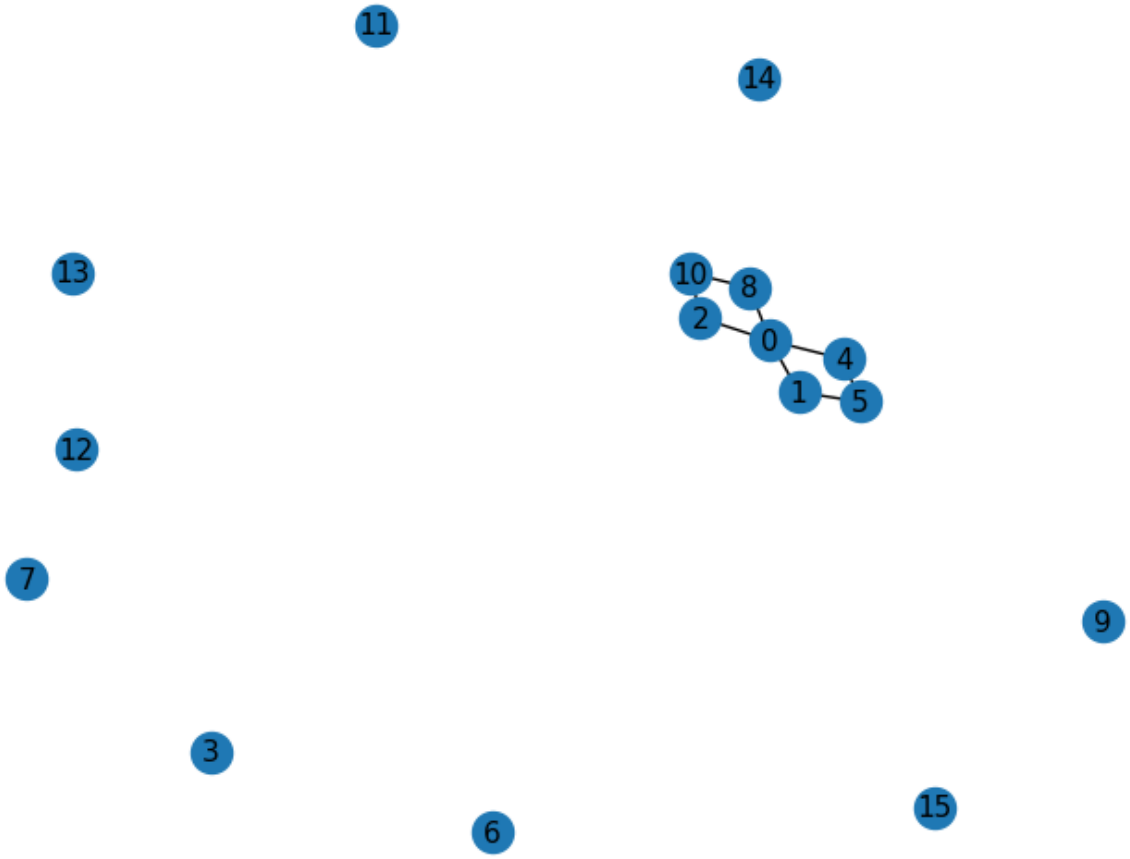


FIG. 2. The adjacency graph of thee Hamiltonian for $L = 4$

[1] OEIS Foundation Inc., The Lucas Numbers, Entry A000032, *The On-Line Encyclopedia of Integer Sequences* (2026), accessed: 2026-06-16.

[2] D. J. Luitz, Computational methods in condensed matter theory, Lecture notes for physics767, University of Bonn.

[3] U. Schollwöck, The density-matrix renormalization group in the age of matrix product states, Annals of Physics **326**, 96 (2011), arXiv:1008.3477 [cond-mat.str-el].